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***B.Tech. Degree V Semester Special Supplementary Examination
August 2015***

**ME 1503 MACHINE TOOLS AND MACHINING SCIENCE
(2012 Scheme)**

Time : 3 Hours

Maximum Marks : 100

(Draw diagrams wherever necessary)

PART A(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Write briefly on Capstan and Turret lathes bringing out their differences.
 (b) What is the difference between drilling and boring operations? What types of tools are used in these operations?
 (c) What is indexing in a milling machine?
 (d) Write short notes on:
 (i) Lapping (ii) Honing
 (e) Explain with the help of diagrams the tool signature of a single point tool.
 (f) Differentiate between orthogonal and oblique cutting.
 (g) What do you understand by shear plane in machining? Write down the Merchant's and Lee & Shaffer's solutions for the determination of shear angle.
 (h) Define tool life. Write a note on any two machinability criterion.

PART B

(4 × 15 = 60)

- II. (a) Explain the various work holding devices used in a conventional centre lathe, with neat sketches. (7)
 (b) Explain various methods available for taper turning in a lathe with specific advantages and limitations. (8)
- OR**
- III. (a) Explain the operation and need for a clapper box in a mechanical shaper, with neat sketch. (5)
 (b) Explain the construction of a radial drilling machine with emphasis on how the requisite motions are obtained. (10)
- IV. (a) Differentiate between Upmilling and Downmilling process. (5)
 (b) Explain the construction of dividing head used in milling operations, with neat sketches. (10)
- OR**
- V. (a) What are the different types of grinding operations? Explain with neat sketches. (10)
 (b) What are the advantages and limitations of using centre less grinding? (5)

(P.T.O.)

- VI. (a) Show the mechanism of chip formation in metal cutting. (7)
(b) Explain the mechanism of built-up edge formation in machining. What are the methods to reduce it? (8)

OR

- VII. (a) Which are the areas of heat generation in machining? (5)
(b) Explain with the help of a graph the percentage of heat distribution at different components involved in machining. (10)

- VIII. (a) Explain flank wear and crater wear in tools, with the help of diagrams. (8)
(b) What do you understand by economics in machining? (7)

OR

- IX. (a) Draw the Merchant's circle diagram and explain the different forces involved in machining. (6)
(b) What are the different types of rake angles used in tools and where these tools are used? (9)
